ATSS — Home Test Shopping List

Anti-Theft Smart System | Minimum Hardware for End-to-End Testing

Goal

Test the full pipeline at home (camera detection + alerts on phone) before investing in outdoor hardware. Everything else runs on your existing laptop, phone, and home WiFi.

1. What to Buy

	#	ITEM	EXACT SPECIFICATION	EST. COST
☐	1	**Orange Pi Zero 3**	2GB RAM variant, Allwinner H618, USB-C power input	**2,500 - 3,500**
☐	2	**MicroSD Card**	SanDisk / Samsung, 32GB, Class 10, A1 rated	**250**
☐	3	**USB Webcam**	Any UVC-compliant USB camera (see camera options below)	**400 - 1,800**
☐	4	**Micro-HDMI to HDMI**	Cable or adapter. Needed for first boot display output on Orange Pi.	**150**
		TOTAL		**~ 3,400 - 4,500**

Amazon / Online Search Terms

Orange Pi Zero 3 2GB

Where to buy (check prices — they vary):
- **Amazon.in** — search "Orange Pi Zero 3 2GB" (price hidden, add to cart to see)
- **Probots.co.in** — Indian SBC retailer, 1-3 day shipping
- **Zbotic.in** — stocks Orange Pi boards
- **CrazyPi.com** — Indian Orange Pi dealer
- **AliExpress** — cheapest (~$20) but 7-15 day shipping from China

SanDisk 32GB MicroSD A1 | **USB webcam Linux UVC 720p** | **Micro HDMI to HDMI cable**

Camera Options — Pick One

OPTION	MODEL	RESOLUTION	FOV	LINUX	PRICE	BEST FOR
A	**Zebronics Zeb-Crystal Clear**	480p @ 30fps	~60°	USB UVC	**~400**	Cheapest option to validate pipeline
B	**Logitech C270 HD**	720p @ 30fps	60°	USB UVC	**~1,100 - 1,800**	Best quality for home test (recommended)

C	**InnoMaker 720P UVC Camera**	720p @ 30fps	120°	USB UVC	**~1,200 - 1,800**	Wide angle, Raspberry Pi compatible, compact module
D	**InnoMaker 1080P UVC Camera**	1080p @ 30fps	130°	USB UVC	**~1,500 - 2,200**	Best for reuse in final outdoor build
E	**Arducam 5MP OV5648**	720p @ 30fps	Wide	USB UVC	**~1,500 - 2,500**	High quality sensor, compact bare module
F	**ELP 170° Fisheye NoIR USB**	720p @ 30fps	170°	USB UVC	**~600 - 1,200**	Final outdoor camera — buy this if you want to reuse later

Recommendation

Tight budget? Buy Option A (Zebronics, ~Rs.400) — cheapest way to validate the full pipeline works on ARM.
Want quality? Buy Option B (Logitech C270, ~Rs.1,100) — best Linux support, well-tested, 720p.
Want to reuse for farm? Buy Option F (ELP 170° NoIR) — this is the actual camera for the final build. Wide-angle + no IR filter for night vision later.

Amazon.in Search Links

Zebronics Zeb-Crystal Clear webcam **Logitech C270 HD webcam** **InnoMaker 720P USB UVC camera**
ELP 170 degree USB camera NoIR

Before buying the camera

Any USB webcam works for home testing. You do NOT need the special wide-angle NoIR camera yet. Even a Rs.400 basic webcam is fine. The NoIR + wide-angle camera is only needed for outdoor night vision later. Look for "UVC", "Plug and Play", or "No driver needed" in the listing.

2. What NOT to Buy (Yet)

ITEM	WHY SKIP FOR HOME TEST	BUY LATER WHEN
~~Second camera~~	~~Test detection accuracy with one first~~	~~After single-cam works reliably~~
~~4G USB dongle + SIM~~	~~Home WiFi connects Pi to laptop~~	~~Before farm deployment~~
~~Powered USB hub~~	~~One camera + WiFi = Pi handles power fine~~	~~When adding 2nd camera + 4G dongle~~
~~Power bank (UPS)~~	~~Home power is stable~~	~~Before farm deployment~~
~~IR LED ring~~	~~Test in daylight first~~	~~Before outdoor night testing~~
~~Tamper sensor (SW-420)~~	~~Software mock works for testing~~	~~During final assembly~~

~~Fan, enclosure, pole~~	~~Not needed for desk testing~~	~~During final assembly~~
~~Surge protector~~	~~Home power is stable~~	~~Before farm deployment~~

3. Use What You Already Have

	ITEM	PURPOSE IN TEST SETUP
☐	**Laptop**	Runs backend server (Docker: Postgres + Mosquitto MQTT + MinIO + FastAPI)
☐	**Phone** (Android)	Flutter mobile app — receives alerts, controls device
☐	**Home WiFi**	Connects all three devices on the same network
☐	**USB keyboard**	Orange Pi first boot setup (one-time, then use SSH)
☐	**TV** / **Monitor** (HDMI)	Orange Pi first boot display (one-time, then use SSH)
☐	**USB-C phone charger** (5V/2A+)	Powers the Orange Pi (any decent phone charger works)

4. Home Test Architecture

```
                        Home WiFi (192.168.x.x)
                               |
              +----------------+----------------+
              |                |                |
           LAPTOP          ORANGE PI          PHONE
        192.168.x.10      192.168.x.20      192.168.x.30
       +-------------+      +------------+      +----------+
       | Docker:     |      | Edge       |      | Flutter  |
       |  Postgres   |<---->| Daemon     |      | App      |
       |  Mosquitto  | MQTT |            |      |          |
       |  MinIO      |      | + USB      |      | Receives |
       |  FastAPI    |<---->| Camera     |<---->| alerts + |
       |  Backend    | WiFi |            | WiFi | images   |
       +-------------+      +------------+      +----------+

       Flow: Camera --> Motion Detect --> YOLO --> MQTT Alert --> Backend --> Phone
```

5. Day-by-Day Setup Plan

Day 1 — Orange Pi Setup

1. Download Armbian Bookworm image for Orange Pi Zero 3 from **armbian.com**
2. Flash to MicroSD using **Etcher** (balenaEtcher) on laptop
3. Insert MicroSD, connect micro-HDMI to TV, USB keyboard, and USB-C power
4. First boot: set root password, create user account
5. Connect to WiFi: `sudo nmtui`
6. Note IP address: `ip addr show wlan0`
7. SSH from laptop: `ssh user@192.168.x.x` — disconnect HDMI and keyboard

Day 2 — Install Software on Pi

```
sudo apt update && sudo apt install -y python3 python3-venv python3-pip git v4l-utils
```

```
git clone <your-repo-url> /opt/atss
cd /opt/atss/edge
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt
bash download_model.sh
```

If onnxruntime fails to install

Try: `pip install onnxruntime==1.16.3`

If still fails: `sudo apt install python3-opencv` and try older onnxruntime versions.

This is the #1 blocker — don't proceed until this works.

! Verify: `python3 -c "import cv2; import onnxruntime; print('OK')"` must print **OK**

Day 3 — End-to-End Test

On Laptop

```
cd anti-theft-smart-system/backend
docker-compose up -d
curl http://localhost:8000/health    # should return {"status": "ok"}
```

On Orange Pi

```
# Plug in USB camera, then:
v4l2-ctl --list-devices
# Quick test:
python3 -c "import cv2; c=cv2.VideoCapture(0); print(c.read()[0])"
# Should print: True
```

```
# Edit config — set MQTT broker to laptop IP:
cp config/production_config.yaml config/device_config.yaml
nano config/device_config.yaml
#   mode: "production"
#   mqtt.broker: "192.168.x.10"    (your laptop IP)
#   camera.production.cam1_source: "/dev/video0"
#   camera.production.cam2_source: "/dev/video0"
```

```
# Run the edge daemon:
cd /opt/atss/edge && source venv/bin/activate
python3 -m src.main --config config/device_config.yaml --debug
```

On Phone

1. Open Flutter app > Settings > set server URL to `http://192.168.x.10:8000`
2. Register a new account > Login
3. Link device with UID: `FARM-001`
4. Walk in front of the camera
5. Check phone — alert with your snapshot should appear!

6. It Works When...

- [ ] Pi terminal shows: "Motion detected > YOLO inference > person detected > alert published"
- [ ] Laptop Mosquitto logs show: message on `farm/farm_1/device/FARM-001/alert`
- [ ] Phone app shows: new alert card with snapshot image of you
- [ ] Time from walking in front of camera to alert on phone: under 10 seconds
- [ ] System runs for 1+ hour without crashing or memory issues

- [ ] Arm/disarm from phone app actually starts/stops detection on Pi

7. Cost Comparison

	HOME TEST (NOW)	FULL FARM BUILD (LATER)
Orange Pi Zero 3 (2GB)	2,500 - 3,500	2,500 - 3,500
MicroSD 32GB	250	250
USB Camera (1x basic / 2x NoIR)	500	1,200
Micro-HDMI cable	150	150
IR LED Ring x2	-	400
4G USB Dongle	-	600
SIM Card + Plan	-	150/mo
Powered USB Hub	-	200
5V/3A Adapter + Surge Protector	-	350
Power Bank 10000mAh (UPS)	-	500
Tamper Sensor (SW-420)	-	30
Fan + Enclosure + Mounting	-	580
Cables, Glands, Ties	-	350
TOTAL	**~ 3,400 - 4,500**	**~ 7,200 - 8,300**

Everything you buy now carries forward

The Orange Pi, MicroSD, and micro-HDMI cable are reused in the final build. The basic webcam is the only throwaway (~Rs.500) — replaced by NoIR cameras for outdoor use. **You risk only Rs.500 if the project doesn't work out on ARM.**

8. Important Notes

Orange Pi Zero 3 — Not Raspberry Pi

The Orange Pi Zero 3 looks similar but is NOT a Raspberry Pi. It uses a different SoC (Allwinner H618). Do NOT buy Raspberry Pi accessories (cases, hats) — they won't fit. Make sure you get the **Zero 3** model specifically, not Zero, Zero 2, or Zero 3B.

MicroSD Quality Matters

Buy a genuine SanDisk or Samsung card. Counterfeit cards are common and will corrupt within weeks of 24/7 operation. Buy from Amazon (sold by Appario/Cloudtail) or an authorized retailer.

Camera Compatibility Check

Before buying, check if the camera says "UVC compatible" or "driverless" or "works with Linux." Any webcam made after 2015 from brands like Logitech, Zebronics, or iBall should work. Avoid cameras that say "Windows only" or require driver installation.

Delivery Times

Orange Pi Zero 3 may take 5-10 days if ordered from AliExpress. Check Amazon.in or Robu.in for faster delivery (2-3 days). MicroSD and webcam are available at any local electronics shop — buy same day.